

Matematyka Dyskretna

Cwiczenia 1 - Rozwiązania

① $U = \{*, \square, 1, \blacksquare, \circ, \pi, \heartsuit, \spadesuit\}$ $C = \{*, \circ\}$
 $B = \{\heartsuit, \square, \pi\}$ oraz $A = \{*, \square, 1, \blacksquare, \circ\}$

(i) $A \cup B = \{*, \square, 1, \blacksquare, \circ, \heartsuit, \pi\}$ (ii) $A \cap B = \{\square\}$

$A \setminus B = \{*, 1, \blacksquare, \circ\}$ $B \setminus A = \{\heartsuit, \pi\}$

Zauważ że $A \setminus B \neq B \setminus A$. Ale zawsze

$A \cup B = B \cup A$ i $A \cap B = B \cap A$

$A' = \{\pi, \heartsuit, \spadesuit\}$

$B' = \{*, 1, \blacksquare, \circ, \spadesuit\}$

"
 $U \setminus A$

"
 $U \setminus B$

$A \oplus B = (A \cup B) \setminus (A \cap B) \stackrel{(i) \text{ i } (ii)}{=} \{*, 1, \blacksquare, \circ, \heartsuit, \pi\}$

$C \subseteq A$ bo każdy element z C jest też w A
 tak

$C \subseteq B$ nie bo np $*$ $\in C$ ale $*$ $\notin B$

$C \times B = \{(*, \heartsuit), (*, \square), (*, \pi), (\circ, \heartsuit), (\circ, \square), (\circ, \pi)\}$

$B \times C = \{(\heartsuit, *), (\heartsuit, \circ), (\square, *), (\square, \circ), (\pi, *), (\pi, \circ)\}$
 $P(\emptyset) = \{\emptyset\}$

Widac, że $C \times B \neq B \times C$ 2-el. podzbiory

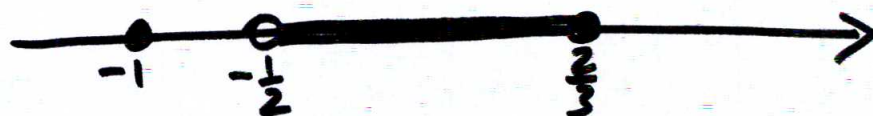
$P(B) = \{\emptyset, \{\heartsuit\}, \{\square\}, \{\pi\}, \{\heartsuit, \square\}, \{\heartsuit, \pi\}, \{\square, \pi\}, \{\heartsuit, \square, \pi\}\}$

zero el. podzbiory 1-el. podzbiory 3-el. podzbiory

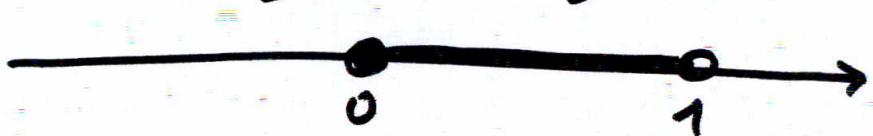
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CWICZENIA 1 - ROZWIĄZANIA

A na osi $\mathbb{R}$



B na osi $\mathbb{R}$



$$A \cup B = \{-1\} \cup (-\frac{1}{2}, 1) \quad A \cap B = [0, \frac{2}{3}]$$

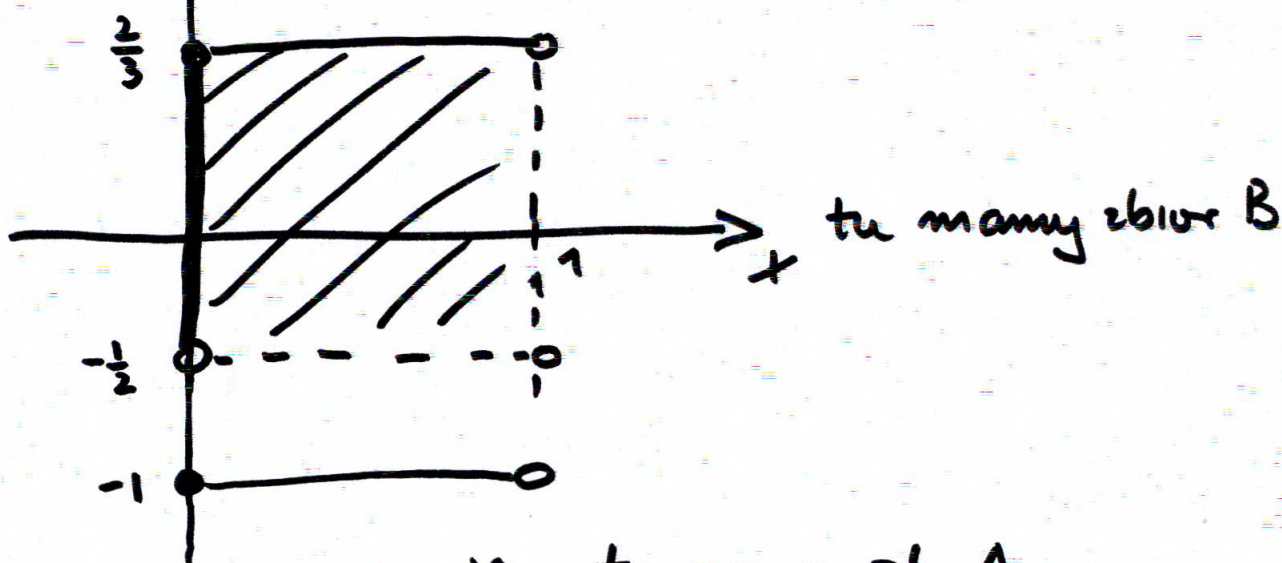
$$A \setminus B = \{-1\} \cup (-\frac{1}{2}, 0) \quad B \setminus A = (\frac{2}{3}, 1)$$

$$A' = (-\infty, -1) \cup (-1, -\frac{1}{2}] \cup (\frac{2}{3}, +\infty)$$

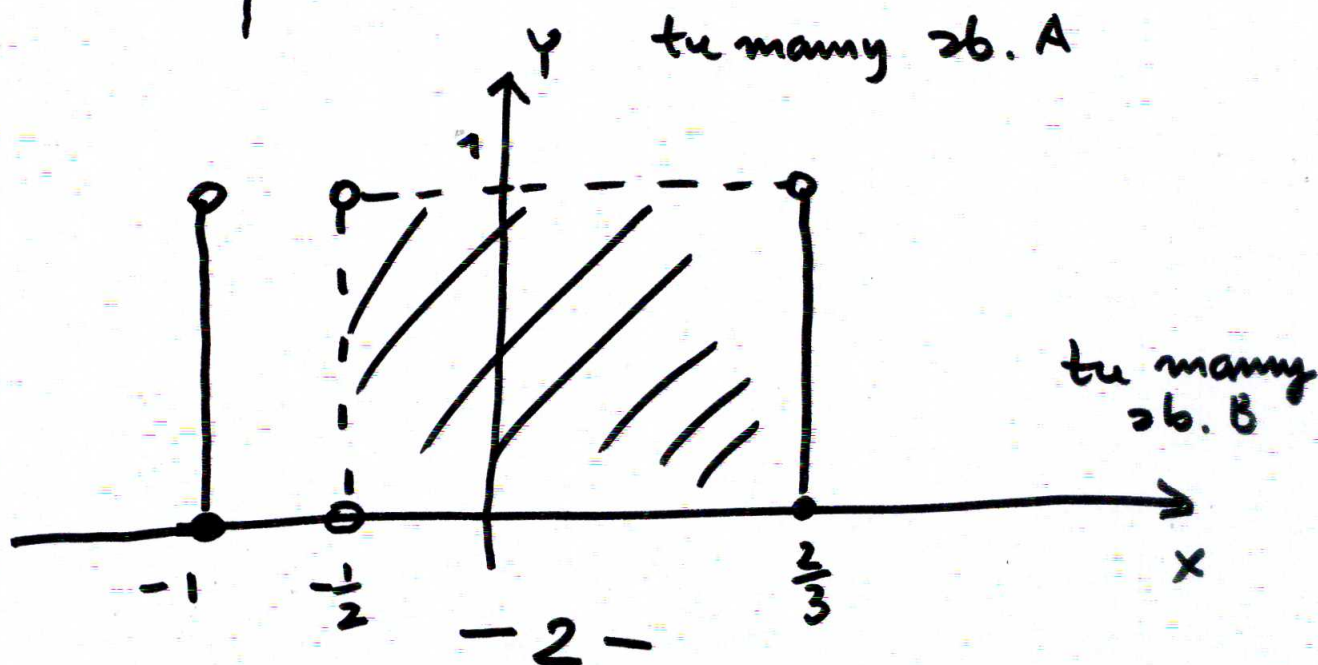
$$B' = (-\infty, 0) \cup [1, +\infty)$$

y tu mamy zbiór A

$B \times A$



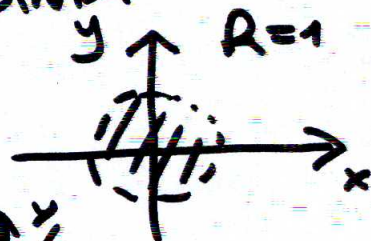
$A \times B$



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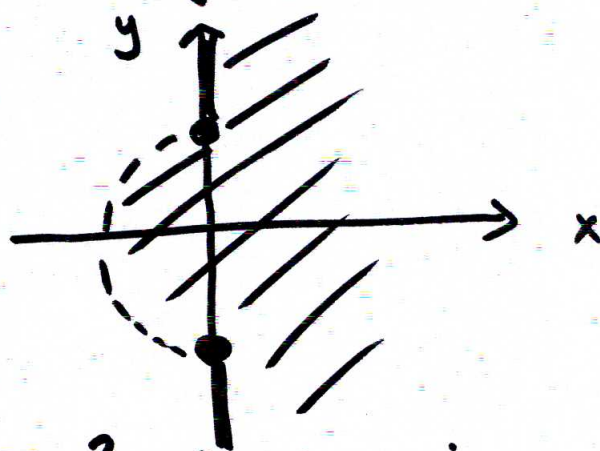
③ $A = \{(x, y) : x^2 + y^2 \leq 1\}$



$B = \{(x, y) : x \geq 0\}$

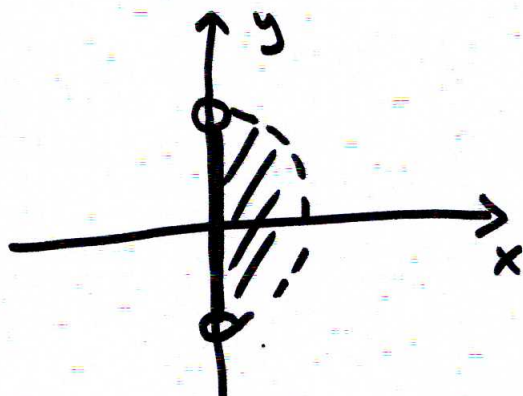


$A \cup B = \{(x, y) : x^2 + y^2 < 1 \text{ lub } x \geq 0\}$ analitycznie



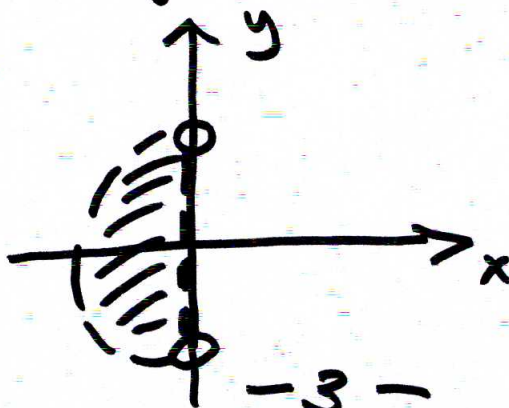
$A \cup B$
geometrycznie

$A \cap B = \{(x, y) : x^2 + y^2 < 1 \text{ i } x \geq 0\}$ analitycznie



$A \cap B$
geometrycznie

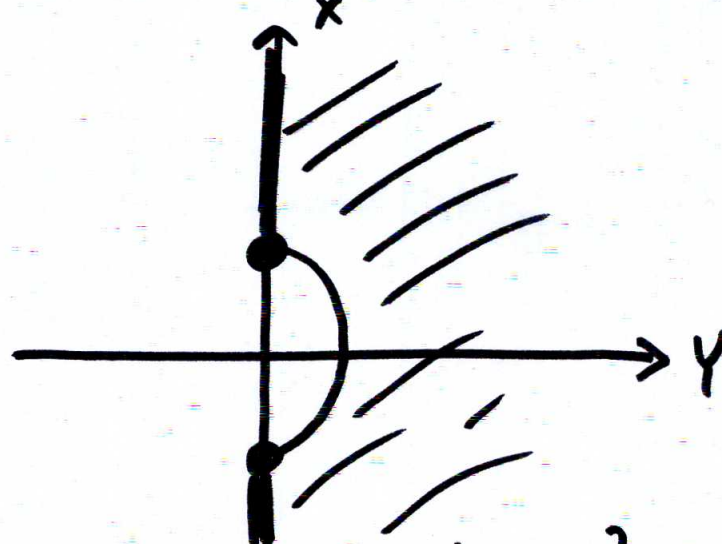
$A \setminus B = \{(x, y) : x^2 + y^2 < 1 \text{ i } x < 0\}$ analitycznie



$A \setminus B$

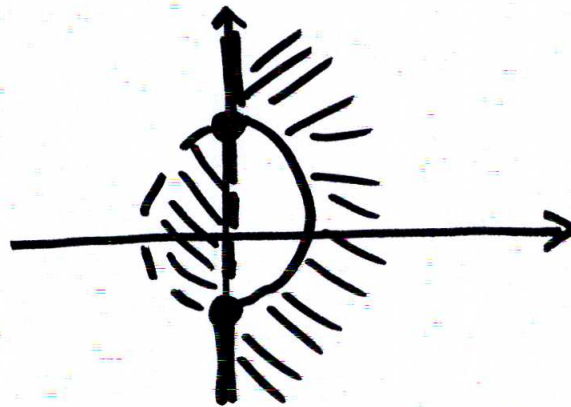
MATEMATYKA DYSKRETNA ĆWICZENIA 1 - ROZWIĄZANIA

$B \setminus A$



$$B \setminus A = \{(x, y) : x \geq 0 \text{ i } x^2 + y^2 \geq 1\}$$

$A \cup B$



Nie da się zrealizować $A \times B$ bo
 A jest dwuwymiarowe i B też $\Rightarrow$
 $A \times B$ jest czterowymiarowe.